Malware Architecture & Cybersecurity Quick Recall Notes

COMPREHENSIVE GUIDE • STUDENT ID: 24IT009

Aa evu software chhe je system ne **damage kare**, data **steal kare**, **spying kare**, **unauthorized access** ape, ane files ne **encrypt kare**. Cyber attackers malware no use kare chhe systems compromise karva mate.

Malware Structure Hierarchy

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Malware
├── Virus
├── Worm
├── Trojan
├── Spyware
├── Ransomware
├── Adware
├── Rootkit
├── Keylogger
├── Botnet Malware
└── Fileless Malware
```

1. Virus

What is Virus?

Computer virus ek evu malicious code chhe je potani **own copy banave** chhe, target **files infect kare** chhe, ane systems ma **spread thay** chhe. Jem human virus ek person thi bija person ma faelay, temaj aa computer ma spread thay chhe.

How Virus Works?

Infected File → **User Open Kare** → **Virus Execute Thay** → **Other Files Infect Kare**

Virus Characteristics

Feature	Description
Self-replication	Potani **own copy banave** chhe.
File infection	Existing healthy **files infect kare** chhe.
User action required	Execution mate mostly **file open karvu pade** chhe (No automatic activation).
Slow spread	User interaction ane sharing thouth j spread thay chhe.

Types of Virus

Type	Work / Operating Mechanism
File Injector	**Executable files (.exe) infect** kare chhe.
Boot Sector Virus	System ni **Boot process attack** kare chhe ane MBR damage kare chhe.
Macro Virus	Word, Excel documents ane templates ne infect kare chhe.
Polymorphic Virus	Detection avoid karva mate potanu **shape/code change kare** chhe.
Resident Virus	Computer ni **Memory (RAM) ma active** rahe chhe.

Famous Example: **ILOVEYOU virus** – Je malcious email attachment throug akhkha world ma spread thayo hato.

Prevention Layers: Antivirus install rakho • Unknown files avoid karo • Regular software updates karo • Email attachments open karti vakhate caution rakho.

2. Worm

What is Worm?

Worm ek standalone **self-replicating malware** chhe je **automatically spread thay** chhe. Tene spread thava mate koi human active interaction ni jarur nathi hoti, te akha **network through propagate** kare chhe. A j karne te Virus karta **vadhu dangerous** manvama ave chhe.

How Worm Works?

One Computer Infect → Network Scan → Other Systems Infect → **Automatic Rapid Spread**

Worm Characteristics

Feature	Description
Self-spreading	Vina koi help automatic spread thava sakhsham chhe.
No user action	Tene execute thava mate **user interaction nai joiye**.
Network-based	Network vulnerabilities no use kari connectivity thi spread thay.
Fast spreading	Extremely fast speed thi pooray network ne down kari shake chhe.

Famous Example: **WannaCry ransomware attack** ma pan network propagation mate SMB vulnerability exploit karva worm mechanism no use thayo hato.

Prevention Layers: Network Firewall active rakho • Regular Patch updates apply karo • Disable unnecessary services • IDS/IPS (Intrusion Detection/Prevention Systems) deploy karo.

3. Trojan

What is Trojan?

Trojan athva Trojan Horse ek **fake legitimate software** chhe. User ne dekhva ma lage ke aa ek ✅ **normal app** chhe, pan reality ma te ek ❌ **malicious software** hoy chhe je system pachal thi control kare chhe.

How Trojan Works?

Fake Software downloaded → **User Install Kare** → **Backdoor Create Thay** → **Attacker Access Male**

Trojan Characteristics

Feature	Description
Disguised	Useful or legit software jevu dhong kare chhe.
No self-replication	Te potani aape spread nathi thto, user bhulthi download kare chhe.
Backdoor create	Attacker ne system no hidden **unauthorized access** ape chhe.
Data theft	Sensitive credential information steal kare chhe.

Types of Trojan

Type	Work / Operation
Backdoor Trojan	Attacker ne system no remote access provide kare.
Banking Trojan	User na confidential **Bank data steal** kare chhe.
Downloader Trojan	Target system ma badama badu **other malware install** kare chhe.
RAT Trojan	Remote Access Trojan - Akha system nu remote control generate kare.

Famous Example: **Zeus malware** – Je main target secure banking credentials steal karva mate badnaama thayo hato.

Prevention Layers: Target strictly verified and trusted downloads • Strong Antivirus protocol • Crack software completely avoid karo • Unknown email attachment caution.

4. Spyware

What is Spyware?

Spyware ek gupt malicious software chhe je user ni jany baahar secretly system ne **monitor kare** chhe, crucial **user data collect kare** chhe, ane unki **browsing habits track kare** chhe. Te completely **without permission** kaam kare chhe.

How Spyware Works?

Silent Installation → **User Activity Track** → **Passwords & Data Collect** → **Attacker Ne Mokle**

Spyware Characteristics

Feature	Description
Hidden	System na background ma khangi rite chale chhe.
Monitoring	Continuous active user activity track kare chhe.
Data theft	Credentials, personal information collect kare chhe.
Privacy violation	User upar continuous spying thava thi privacy secure nathi rehti.

Types of Spyware

Type	Work / Impact
Keylogger	User na direct **Keyboard strokes record** kare chhe (passwords chore).
Tracking Cookies	Web browsing activity ane history closely track kare.
Adware Spyware	Unwanted Ads show kare ane tracking pan sathe chalu rakhe.
System Monitor	Full screen uploads ane detailed full activity monitor kare.

Prevention Layers: Dedicated Anti-spyware tools deploy karo • High browser security settings • Avoid suspicious software downloads • OS/App updates.

5. Ransomware

What is Ransomware?

Ransomware ek atyant khatarnak attack chhe je user ni badhi critical **files encrypt kare** chhe, pooray system no **access block kare** chhe, ane unhe unlock karva mate moti **ransom demand (खंडनी)** mange chhe. Transacation hide rakha mate payment mostly **cryptocurrency (Bitcoin)** ma mange chhe.

How Ransomware Works?

System Infect → **Files Encrypt (Lock)** → **Access Blocked** → **Ransom Note Displayed**

Ransomware Characteristics

Feature	Description
Encryption	Cryptographic algorithms thi crucial **files lock kare** chhe.
Ransom demand	Decryption key badle heavily money mange chhe.
Data hostage	Files user mate purepoori inaccessible bani jay chhe.
Financial damage	Business ane individuals ne huge financial losses khamva pade.

Types of Ransomware

Type	Work / Threat Vector
Crypto Ransomware	Main target data storage standard **files encrypt** kare chhe.
Locker Ransomware	Akhkhe **Device no main access lock** kari de chhe.
Double Extortion	Data return na badle explicit internet par **Data leak karvani dhamki** ape.
RaaS	Ransomware-as-a-Service (Affiliate business module for attackers).

Famous Example: **WannaCry Ransomware** – Je 2017 ma akha world ma failayelo mota paaye industrial disruption karva mate jawabdari hato.

Prevention Layers: Regularly maintain **Offline backups** • Robust Antivirus & EDR • Patch management • Strict Email awareness training • Network segmentation.

Core Comparison Matrix

Malware Type	Spread Method	Main Purpose	User Action Needed?
Virus	File-to-file attaching	Damage / infect files	✅ **Yes** (File open karvu pade)
Worm	Network Vulnerabilities	Rapid automatic network spread	❌ **No** (Self-spread)
Trojan	Fake legitimate software	Create Backdoor / data theft	✅ **Yes** (Bhulthi install kare)
Spyware	Silent bundled software	Hidden monitoring / steal info	Sometimes
Ransomware	Multiple methods (Phishing/ Worms)	Encrypt files & demand money	Usually yes (to initiate)

Real-Life Impact Summary

Malware	Real-Life Damage / Core Consequence
Virus	Severe File corruption and operating system failure
Worm	Massive Network shutdown and bandwidth exhaustion
Trojan	Total Unauthorized root access to malicious third-parties
Spyware	Critical Privacy theft, banking credential loss, identity fraud
Ransomware	Total Business disruption, heavy monetary losses, asset leakage

Malware Defense Layers Architecture

Antivirus → **Firewall** → **IDS/IPS** → **Patch Mgmt** → **Backups** → **Awareness** → **SOC**

Cybersecurity Domains Related

- **Defensive Security:** Infrastructure protection, hardening, and control deployment.
- **SOC (Security Operations Center):** Continuous real-time infrastructure **Monitoring**.
- **Incident Response:** Live threat suppression and immediate post-attack handling.
- **Digital Forensics:** Post-mortem deep technical investigation into how breach occurred.
- **Threat Hunting:** Proactive identification and hunting down hidden malware indicators.

FINAL SUMMARY TABLE FOR RAPID RECALL

Malware malicious software chhe je systems, networks ane users ne damage, spying, data theft athva financial loss karva mate design karvama aave chhe. Virus, Worm, Trojan, Spyware ane Ransomware tena major types chhe je alag-alag techniques thi systems compromise kare chhe. Aa document tamari quick exam/interview preparation ma rapid recall mate useful thashe.